

BRIAN ZHOU LIU

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EDUCATION

University of California, San Diego

B.S. in Data Science (AI & ML Specialization)

La Jolla, California

GPA: 3.95/4.00 | Expected June 2028

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript/JavaScript, Rust, Bash

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, React, FastAPI

Data & Infrastructure: PostgreSQL, MongoDB, ChromaDB, Git, Docker, GCP (Cloud Run, Cloud SQL)

EXPERIENCE

Q-Lab, UC San Diego

La Jolla, California

Research Intern (Advisor: Prof. Lianhui Qin)

January 2026 - Present

- Co-developed SIGA, a Claude Code adapter that automatically configures scientific simulators; built its ChromaDB retrieval layer, MCP XML validator, and plugin framework, and co-authored the accompanying preprint ([arXiv:2606.09774](https://arxiv.org/abs/2606.09774)).
- Led a 30-task OpenFOAM transfer study; SIGA's best configuration scored 0.870 accuracy with 30/30 complete cases.
- On harder held-out GEOS tasks, SIGA **raised accuracy by 9.6% (0.720 to 0.789) and reduced run-to-run standard deviation by ~16x**.
- Developed a direct encoder-decoder Transformer to forecast four future single-cell states from 100,000 trajectories; achieved $R^2 = 0.536$ across 3,201 output genes and **$R^2 = 0.857$ on the top 50 dynamic genes**.

Asakana (YC F26)

Remote

Product Development Intern

October 2025 - February 2026

- Deployed a first-generation OCR/ETL pipeline with Gemini Flash-Lite, **automating previously manual supplier entry for 10,000+ products** from PDF/Excel sheets into MongoDB.
- Developed a Dialogflow CX ordering agent backed by Cloud SQL and REST APIs, with Twilio SMS notifications and automated pricing-rule enforcement.

Rare AI Lab, UC San Diego

La Jolla, California

Research Intern (Advisor: Prof. Aobo Li)

September 2024 - December 2025

- Implemented a multi-fidelity surrogate optimizer combining conditional neural processes and Gaussian processes, reducing detector-simulation cost by 90% for experimental design exploration.
- Co-first-authored a paper accepted to the NeurIPS 2025 ML4PS Workshop:** "Efficient Optimization of COHERENT Detector Design Parameters with RESuM."

MAIX Lab, Emory University

Atlanta, Georgia

Research Intern | Regeneron STS Top 300 Scholar (Advisor: Prof. Ran Xiao)

March 2023 - November 2023

- Constructed anatomically informed feature tensor for cardiac screening using XResNet, **achieving 93.7% AUC**.
- Boosted model **sensitivity to 85.5%** by implementing with 1D convolutional layers that capture cross-lead patterns.

SELECTED PROJECTS

FlashMath: AI-Assisted Math Study Desktop App

Independent Project | [GitHub](#)

Tauri, Rust, React, TypeScript

2026

- Completed the MVP for a cross-platform Tauri desktop app with Rust/React, local persistence, screenshot/PDF ingestion, LaTeX rendering, and provider-agnostic AI OCR.
- Implemented a modified SM-2 algorithm that incorporates review time and compresses review intervals around exam deadlines.

Blueprint: Codebase Summarization & Vulnerability Detection

Pittsburgh, Pennsylvania | [Devpost](#)

NexHacks | 1st Place DevTools

January 2026

- Built the AST parsing pipeline and VS Code UI to extract repository structure and render interactive React Flow/D3.js dependency graphs inside the editor.
- Integrated bear-1/Gemini and parallel directory-level agents for summarization and security analysis; demoed on repositories as large as MongoDB and won 1st Place (DevTools) at NexHacks.

CARP: Health Monitoring & Catch Optimization System

Stanford, California | [Slides](#)

Sushi Hackathon at Stanford | 3rd Place

October 2025

- Built an Arduino brace with FSR-402 pressure sensing and Bluetooth telemetry; integrated a Random Forest trained on age, BMI, and clinical features into a Python dashboard with live monitoring, risk scores, alerts, and historical trends.
- Designed a full-stack dashboard featuring an AI agent with RAG for market insights, a health module to track CTS risk using the brace's data, and visualizations of a 50,000+ point dataset targeting optimal geospatial fishing locations.